A Body Biasing Method For Charge Recovery Circuits: Improving The Energy Efficiency and DPA-Immunity

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***Abstract*—Charge recovery is a promising concept to design (cryptographic) VLSI circuits with low energy dissipation. However, unsatisfactory designs of proposed logic cells degrade its theoretical efficiency significantly both in its energy consumption and the resistance against differential power analysis attacks (DPA-attacks). Short circuit dissipation and non-adiabatic discharging of capacitance loads are the two major sources of this degradation which are addressed in this paper. In order to reduce these dissipation significantly, we manipulate threshold voltage of circuits transistors by body biasing. To evaluate the efficiency of our method we select a common charge recovery logic called 2N-2N2P and examine it on 8-bit Brent-Kung adder as well as 4-bit, 8-bit and 16-bit 2N-2N2P carry look-ahead adders. Experimental results show at least 50% reduction in the energy consumption as compared to traditional 2N-2N2P. Moreover, using our technique reduces the dynamic power variation by a factor of 7.8 on the 2N-2N2P inverter and therefore improves DPA-resistance of charge recovery circuits significantly.**

I. Introduction

As the complexity of digital circuits grows, the energy consumption has become more critical design factor in digital systems. Several power reduction techniques are utilized to reduce the energy consumption. Among these techniques, in a system with high switching activity, a technique called charge recovery has the potential of dissipating less energy than its counterparts, in a way that several charge recovery logic styles have been proposed during the recent decade [1], [2], [3], [4]. On the other hand CMOS technology is the widely used in cryptographic hardware (*e.g.*, smart cards) in which there are strong threats such as side channel attack (SCA) scenarios [5] and especially differential power analysis (DPA) attacks [6]. Recently, charge recovery styles have been examined as side channel attack countermeasures [7], [8]. Since DPA-attacks, generally, exploit information leakage obtained by analyzing dynamic power to reveal secret keys [6], [9], charge recovery circuits achieve DPA-resistance by significantly decreasing the dynamic power consumption.

Since the dynamic power consumption in charge recovery circuits grows as operational frequency increase; both the energy efficiency and the DPA-resistance of these circuits degrades as frequency of operation increases [7], [8]. This problem is mainly due to the unsatisfactory design of charge recovery circuits [11]. For example, diode-based charge recovery styles have a limited power saving due to the dissipation of the direct current flow through the diodes [1]; also as we will see in section III, latch based styles are usually limited because of a short circuit dissipation that occurs in the beginning of the charging phase [2], [3], [4]. Furthermore, high dynamic power variations due to short circuit dissipation and non-adiabatic discharging leak information out from these styles [7], [8].

A charge recovery style called boost logic has been developed [11] which reduces short circuit dissipation. Although this logic style is efficient in GHz-class charge recovery operations, it cannot operate efficiently at less operating frequencies (*e.g.* few hundred MHz or less), and consequently cannot be used in wide variety of applications such as RFID tags, sensor networks, and several embedded systems. Furthermore, this technique cannot be generalized to other charge recovery logic styles such as ECRL [3], 2N-2N2P [4], PFAL [12].

Body Biasing is an important low power technique which can be simply described by modifying the substrate voltage of the CMOS circuit in order to reduce the leakage dissipation. However, even such a simple idea has complicated aspects so that many researches were done in this area such as [13], [14]. In this paper, we integrate body biasing into charge recovery circuits in order to reduce the short circuit dissipation. To the best of our knowledge none of the previous works have employed body biasing methods in order to reduce the power consumption of charge recovery circuits. For this purpose, we bias the n-substrate of each circuit stage with the stage power clock signal. This reduces the dynamic energy consumption significantly. We select a common charge recovery circuit called 2N-2N2P [4] to examine the proposed technique on 4-bit, 8-bit, and 16-bit 2N-2N2P carry look ahead adders as well as 8-bit Brent-kung adder in different frequencies (between 2.5MHz to 500MHz). The experimental results show at least 50% reduction in energy consumption in comparison to the corresponding traditional charge recovery circuit.

The organization of the paper is as follows. In Section II, the concept of charge recovery logics is illustrated. Then,

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the structure of 2N-2N2P circuits is reviewed briefly. In Section III, our method is described and its characteristics are clarified. Section IV illustrates the simulation results. Also, in this section we compare our proposed method, traditional charge recovery, and static CMOS circuits. Finally Section V concludes our discussions.

II. Charge Recovery Logic

The charging through DC voltage source causes enormous energy dissipation because the charge experiences a potential drop on its way from the supply node to the load. In contrast, in charge recovery circuits each capacitance node is charged steadily, and the voltage drops across the resistive elements are made small in order to reduce the energy dissipation during the charge or discharge of the capacitive loads via a power clock signal.

The principle of the adiabatic charging scheme can be best explained by contrasting it with the conventional method during the charge of a capacitance in an RC circuit. To charge a node with the associated capacitance C from 0 to V_{dd} in conventional CMOS circuits, $Q \cdot V_{dd} = C \cdot {V_{dd}}^2$ is taken from the supply voltage source. Half of it is dissipated in the path resistors, and the rest is stored in the capacitor C. Thus, the energy dissipation in each transition is given by [18]:

$$E_{\text{Conventional}} = \frac{1}{2} C \cdot {V_{dd}}^2. \tag{1}$$

On the other hand, consider a capacitance node of a circuit that is charged by a time-varying voltage source whose slope of transitions is slowed down. In this charging process the overall energy dissipated for each transition is reduced to

$$E_{\text{Adiabatic}} = \xi \frac{R \cdot C}{T} C \cdot {V_{dd}}^2, \tag{2}$$

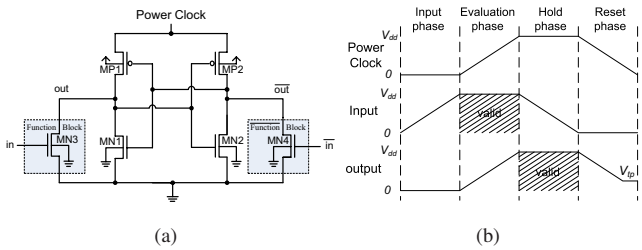
where T denotes the charging/discharging time, V_{dd} is the voltage swing value, and ξ is the shaping factor that supports the other types of voltage source waveform in addition to the ramp waveform. Ideally, the charging energy tends to zero ($E_{\text{Adiabatic}} \rightarrow 0$) by increasing T ($T/RC \rightarrow \infty$). Also, the logic gates should be driven by trapezoidal power clock voltage waveforms in order to achieve the best energy efficiency [10].

A. 2N-2N2P

A 2N-2N2P gate consists of two main parts: (i) two functional blocks whose duty are to construct the gate outputs out and $\overline{\mathrm{out}}$ as shown in figure 1(a). (ii) cross-coupled inverters (sense amplifier) which allow maintaining the right (HI or LO) voltage levels at the outputs.

All 2N-2N2P gates operate at four different phases: input phase, evaluation phase, hold phase, and reset phase. In order to clarify the specification of each phase, the operation of a 2N-2N2P buffer/inverter gate is explained for simplicity.

The basic structure of a 2N-2N2P buffer/inverter gate is shown in figure 1(a). During the input phase, the power clock remains low, and inputs can change. At the end of this phase inputs have taken their own valid values. Suppose that in=HI and $\overline{\mathrm{in}}$=LO; therefore, MN3 is closed, and MN4 is open. In other words, the function block which prepares out signal is closed, and the complementary function block which prepares $\overline{\mathrm{out}}$ signal is open.



(a) (b)

Fig. 1. (a) Structure of a 2N-2N2P buffer/inverter gate (b) Timing diagram in 2N-2N2P logic style

At the start of the evaluation phase both of out and $\overline{\mathrm{out}}$ are LO. As shown in figure 1(b), power clock steadily increases towards HI; thus, $\overline{\mathrm{out}}$ is charged through MP2. In contrast, out node remains LO since it is connected to GND through the function block. Finally, MN1 is closed and outputs are latched by the cross-coupled transistors. Note that during this phase inputs must remain unchanged. In hold phase, inputs switch to LO, and both of the function blocks are switched off. In this situation output values are held by the cross-coupled transistors. Note that during this phase output values are valid and can be used as input at the next stage. Finally, during the reset phase power clock steadily decreases towards LO. By falling power clock, $\overline{\mathrm{out}}$ goes LO via MP2 until it reaches V_{tp0}. Then, MP2 becomes open and $\overline{\mathrm{out}}$ remains unchanged. The remaining charge of $\overline{\mathrm{out}}$ is dissipated non-adiabatically at the next cycle if new inputs cause the complementary function block to switch on. A complex 2N-2N2P gate can be easily implemented by replacing the desired NMOS function block instead of MN3, and the NMOS complementary function block instead of MN4 transistor. More detailed descriptions about 2N-2N2P logic can be found in [4], [7], [8].

III. Proposed Method

Here, we explain the sources of non-adiabatic energy dissipation in the 2N-2N2P. Then we show that this dissipation is proportional to PMOS threshold voltage ($|V_{tp}|$). Also we demonstrate that this dissipation is data dependent. In the second subsection, we reduce $|V_{tp}|$ by using the proposed body biasing. In the third section, we explain the correct form of biasing and its corresponding effects on the DPA resistance level of the charge recovery circuits.

A. Reducing the Non-Adiabatic Dissipation of 2N-2N2P

In 2N-2N2P, there are two major sources of non-adiabatic energy dissipation; (i) energy dissipation related to the partial charge recovery of load capacitances; (ii) energy dissipation related to the short circuit dissipation.

The dissipation related to the partial charge recovery of load capacitances occurs as follows: when the circuit is in the reset

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phase, the charge which has been stored in the load capacitance are recovered through a PMOS transistor, *i.e.,* MP1 or MP2 (in Fig. 1). The charge recovery process is in progress while the voltage of power clock is more than the threshold voltage of the PMOS transistor, V_{tp}. Afterward, PMOS transistors are turned off; therefore the load capacitance charge remains at V_{tp}. The remaining charge will not be recovered and will dissipate the its energy when the corresponding function block (*e.g.,* MN4 in Fig. 1) turns on. Consequently, the dissipated energy is data-dependent and may cause information leakage. The amount of the dissipated energy is given by:

$$E_{\text{load}} = \frac{1}{2} C_{load} \cdot {V_{tp}}^2, \tag{3}$$

where C_{load} is the load capacitance at the output signals and V_{tp} is the threshold voltage of the PMOS transistor.

The second non-adiabatic dissipation is occurred in the beginning of the evaluation phase when V_{pc} is gradually rising from zero. Suppose that in the current cycle, MN3 is OFF and MN4 is ON while in the previous cycle MN3 was ON and MN4 was OFF (and therefore the output capacitances are completely discharged). In the beginning of the evaluation phase, PMOS cross-coupled sense amplifier are OFF (MP1 and MP2 are OFF); thus the leakage current are negligible. Consequently, as the capacitance is charging by a negligible current, the capacitance voltage cannot grow rapidly and retains a negligible voltage value. Rising V_{pc} makes the sub-threshold leakage more and more so that when V_{pc} approaches to the near of V_{tp}, the cross-coupled PMOS transistors gradually are turned on; hence the capacitance of the out (C_{out}) charge up through MP1 (non-adiabatically) while a short circuit dissipation occurs through MN4 and MP2. Since the mentioned dissipation occurs when the inputs changes (the load capacitances are completely discharged), it is also data dependent and may cause information leakage. This dissipation can be approximately computed as follows (For brevity, here we do not provide the proofs):

$$E_{sc} = V_{th} I_{d_0} R_s C_{out} \beta + \frac{1}{2} C_{out} V_{th}^2, \tag{4}$$

where β is a constant, I_{D_0} is current at $V_{GS} = V_{tp}$, V_T is the thermal voltage, η is the inverse slope of the sub-threshold current, R_s is the resistance of PMOS transistor when it is ON, and $V_{th} = |V_{tp}|$. Therefore, considering both Eq.3 and Eq.4, the total non-adiabatic dissipation can be computed by the following equation.

$$E_{dis} = V_{th} I_{d_0} R_s C_{out} \beta + C_{out} V_{th}^2, \tag{5}$$

It can be seen from Eq.5, the energy dissipation in this case decreases by reducing V_{th} ($V_{th} = |V_{tp}|$). V_{tp} is the threshold voltage of cross-coupled PMOS transistors and is equal to:

$$V_{\text{tp}} = V_{tp0} - \gamma(\sqrt{V_{bs} + |2\Phi_F|} - \sqrt{|2\Phi_F|}), \tag{6}$$

which Φ_F is the Fermi potential of the substrate and intrinsic silicon, γ is known as the body effect constant, V_{bs} is the source-bulk voltage of the transistor, and V_{tp0} is the minimum threshold of PMOS transistor. In static CMOS, usually both the source and bulk of the PMOS transistors are connected to a DC power supply, V_{dd}. Therefore V_{bs} becomes zero and consequently $V_{tp} = V_{tp0}$. However, in traditional charge recovery, source of the PMOS transistors are connected to an oscillating voltage (V_{pc}) while the bulk is still connected to V_{dd}. Therefore, V_{bs} varies during the time. Since PMOS transistors switch when $-V_{pc} = V_{tp}$ ($V_{GS} = -V_{pc}$), therefore we have $V_{bs} > 0$ at the switching time. Consequently, regarding Eq.6, $|V_{tp}|$ becomes greater than $|V_{tp0}|$, (*i.e.,* $|V_{tp}| > |V_{tp0}|$).

To decrease $|V_{tp}|$, we bias the PMOS bulk of each circuit stage with corresponding power clock signal. This makes $V_{bs} = 0$ and consequently reduces the threshold voltage of PMOS transistors to about its minimum value V_{tp0}. Thus, the dynamic energy dissipation given by Eq.5 decreases.

B. DPA Immunity of The Biased 2N-2N2P

As mentioned in Section I, when the power dissipation of a circuit is correlated with the circuit data, this makes the circuit vulnerable to DPA-attacks. In order to determine the level of DPA-resistance of 2N-2N2P, we should know how the DPA-attacks work. Generally a DPA-attack strategy consists of the following steps [9]; first, an adversary (attacker) analyzes the cryptographic algorithm and chooses an intermediate result of the cryptographic algorithm (intermediate results are generated by the algorithm during its execution). The intermediate result needs to be a function of a small part of the key which we refer to as k, e.g., 8 bits from a 128-bit key (for more information please refer to [9]). In the second step, the power consumption of the cryptographic device is measured while it encrypts or decrypts several different data blocks (which we refer to as observed power consumption traces). The next step of the attack is to calculate a hypothetical value for every possible choice of k. This is done for each data block whose corresponding power consumption is measured in the previous step. Also, the corresponding hypothetical power consumption values of each data block is estimated for every possible choice of key. In the final step, the hypothetical power consumption values of all possible choices of k are compared with observed power consumptions. The values which have the highest correlation reveal k.

The mutual information metric [15], [16] has been used to determine the amount of information leaked by the logic style (*i.e.,* 2N-2N2P in this case), independently of a particular attack. This metric is given by:

$$\mathrm{I}\left(P_{S_g}; \mathrm{L}^q_{s_g}\right) = \mathrm{H}\left[P_{S_g}\right] - \mathrm{H}\left[P_{S_g} | \mathrm{L}^q_{s_g}\right], \tag{7}$$

where P_{S_g} is a random variable denoting the correct hypothetical power consumption of the target signal (e.g. typically, a part of a secret key) under different queries to the target cryptographic implementation in the correct time (The correct time instant is the moment that the key is being processed and the chosen intermediate result are computed), p_{s_g} is the realization of P_{S_g}, $\mathrm{L}^q_{s_g}$ is a random vector containing the side-channel observations generated by the correct key in the time

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interval $q = [t_1, t_2]$. $\mathrm{l}_{\mathrm{s_g}}^{\mathrm{q=t}}$ is the observed power consumption vector at the time t. Also, $\mathrm{H}[\cdot]$ and $\mathrm{H}[\cdot|\cdot]$ are entropy and conditional entropy respectively.

The maximum mutual information occurs when an adversary guesses the correct key and the correct time instant(s) (*i.e.* $q = t_k$). In this case, Eq.6 can be rewritten as:

$$\mathrm{I}\left(P_{S_g}; \mathrm{L}_{p_{s_g}}^{q=t_k}\right) = \mathrm{H}\left[P_{S_g}\right] - \mathrm{H}\left[P_{S_g}|\mathrm{L}_{p_{s_g}}^{q=t_k}\right]. \quad (8)$$

It should be noted that when the detailed circuit level simulations are used, the hypothetical power consumption values can be an accurate estimation of actual power consumption of cryptographic devices. Therefore we assume these hypothetical power consumption values as the actual one (by the assumption of the correct key guess in the correct time instant). The measurement instrument can be assumed as a Gaussian communication channel. In other words, the attacked device sends the power consumption signal through a *Gaussian channel*; the output of this Gaussian channel is the observed power consumption. It should be noted that the additive Gaussian noises are related to the measurement instrument, power supply noise, quantization noise, and noises related to conducted emissions and radiated emissions [9]. Consequently, the observed power consumption vector can be written as follows (in the correct time instant):

$$\mathrm{l}_{\mathrm{s_g}}^{\mathrm{t_k}} = \mathrm{P}_{S_g} + N_g, \quad (9)$$

where N_g is the additive Gaussian noise. Also, the mutual information of the observed power consumption and the actual power consumption is:

$$\mathrm{I}\left(P_{S_g}; \mathrm{L}_{p_{s_g}}^{t_k}\right) = \mathrm{H}[S_g] - \mathrm{H}[N_g] \leq \frac{1}{2} log(1 + \frac{P}{N}) \quad (10)$$

where P is the power constraint factor of the input which is usually $\frac{1}{n}\sum_{i=0}^{n} p_{s_g} \leq P$ holds for P [17]. The maximum power constraint of the $L_{S_g}^{t_k}$ occurs at the peaks of power consumption. There are two peaks of power consumption in each operating cycle of charge recovery circuits (*e.g.*, 2N-2N2P). The first is related to the short circuit dissipation (at the beginning of the evaluation phase) while the second is for complete discharging of load capacitances [7]. These two peak powers can be calculated as follows (For brevity, here we do not provide the proofs):

$$P_{peak-eval} = V_{th} I_{d_0} + \frac{V_{th}^2}{R_s},$$
$$P_{peak-load} = \frac{V_{th}^2}{R_s}, \quad (11)$$

where R_s is the resistance of turned-on function block (section II-A). According to Eq.11 both peak powers are proportional to V_{tp}. Utilizing the proposed body biasing reduces the PMOS threshold voltage, hence decreases the maximum power constraint as well, and it improves the DPA-immunity.

IV. Experimental Results

In this section, energy consumption of our proposed technique on a four, eight and sixteen bit carry look-ahead adders (CLA) and 8-bit Brent-kung adder are evaluated, and the results are compared with the traditional 2N-2N2P and static CMOS counterparts. HSPICE $0.18\mu\mathrm{m}$ TSMC transistor model with 1.8 V_{dd} supply voltage were used in these experiments and the operating frequencies of circuit were chosen between 2.5 and 500MHz. Since both traditional 2N-2N2P and the proposed method employ the same power clock generator, the energy dissipation of the power clock generator is not considered in the comparison. To estimate short circuit dissipation correctly in static CMOS circuits, all square waved inputs were passed through 10 staged inverter chain to create an input signal with right rise/fall time. The observed fall and rise time are about 40 ps for the aforementioned technology model. We have constructed our test-benches using And/Nand, Or/Nor, Not/Buffer, Xor/Xnor gates. Since our method is a circuit level technique and is applied to all these gates, the effectiveness of our method can be approximately generalized for circuits that are constructed from these logic cells.

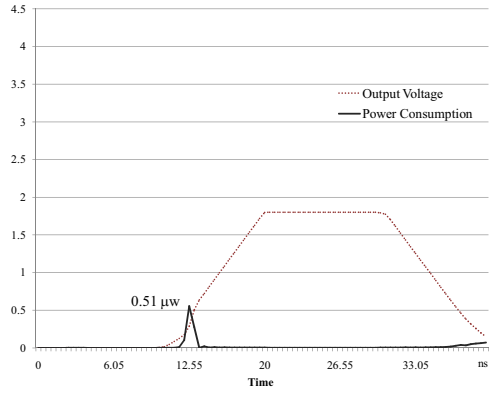
To create HSPICE charge recovery netlists, we synthesized structural Verilog HDL description. The results were converted to HSPICE netlists (by our written program). Afterward, several buffers were inserted into the circuit to adjust circuits with the charge recovery timing constrains (to avoid timing mismatch, etc.).

A. Peak Power Consumption

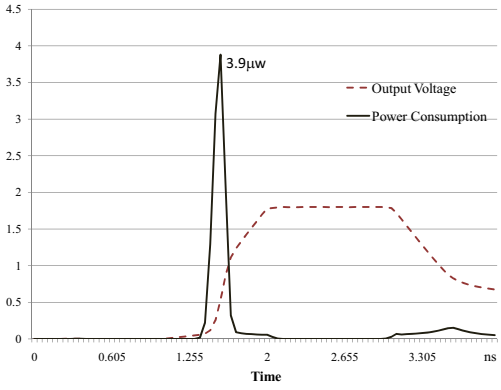
Figure 2 shows the instantaneous power consumption of 2N-2N2P inverter/buffer gate when it operates at 25MHz and drives two 1-fF capacitances. Figure 2(a) shows the output charging process of our proposed body biased 2N-2N2P inverter/buffer. As shown in this figure, in the process of charging the output, a sudden power consumption occurs at about 0.5V. This is because the threshold voltage of the two cross coupled PMOS. At this time, the peak power consumption is about 510nw. On the other hand, figure 2(b) shows the power consumption of a traditional 2N-2N2P inverter/buffer in a condition just the same as the previous. As shown in this figure, the sudden power consumption occurs at about 1.1V. It should be noted that the peak power consumption is 3.9μw which is about 7.8 times greater than when body biasing technique is used.

With respect to DPA resistance, since DPA resistance of 2N-2N2P depends on the capacitances associated with complementary output signals [7], [19], in order to consider the parasitic interconnection capacitances in simulations, two capacitances with amount of $1f\mathrm{F}$ and $10f\mathrm{F}$ have been inserted on each 2N-2N2P gate outputs. Also, since maximum information leaks when maximum data dependent power is consumed, we computed the power constraints only at the peak power, (*i.e.* when the power related to Eq.11 is dissipated). Table I shows the power constraint of NOT/BUFFER, AND/NAND, XOR/XNOR when it operates at 25MHz (using $\frac{1}{n}\sum p_{S_g} \leq P$ for all input transitions). As shown in this

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(a)



(b)

Fig. 2. Peak power consumption when output rises, (a) using our proposed method (b) when no biasing is used. Continuous lines show power consumption while dashed lines show the output voltage waveform

TABLE I
POWER CONSTRAINT OF TRADITIONAL 2N-2N2P VS. THE PROPOSED METHOD.

Gate	NOT/BUFFER	AND/NAND	XOR/XNOR
Traditional(μw)	7.53	6.09	6.37
Biasing(μw)	2.62	1.84	3.34

* Note that less power constraint shows better DPA-immunity (Section III-B).

table, our proposed method reduces the power constraint of the examined gates are reduced significantly (*e.g.* by a factor of 2.87 for NOT/BUFFER, 3.30 for AND/NAND, and 1.97 for XOR/XNOR gates). Consequently, our proposed method shows significant improvement in reducing information leakages. Therefore, our proposed method is more suitable in implementing cryptographic hardware.

B. Energy Consumption

Fig.3(a) through Fig.3(c) represent the energy consumption of a charge recovery 4bit, 8bit, and 16bit carry look ahead (CLA) adder respectively. 4bit CLA adder is made up of 42 two-input gates in addition to 59 inserted buffer/inverters, 8bit CLA adder is made by 180 two-input gates and 325 inserted buffer/inverters, and 16bit CLA adder consists of 1000 two-input gates as well as roughly 2000 inserted buffer/inverter gates. Each node in these figures shows the average energy dissipation caused by random input data. As it can be inferred from Fig.3(a)-Fig.3(c), the energy consumption of circuits when using our proposed biasing are about 50% energy less than their traditional charge recovery counterpart. Also, it can be seen that the energy saving factor of charge recovery regarding the static CMOS counterpart in 16bit CLA adder is less than the previous test cases (as shown in Fig.3(c)). This is due to the vast number of inserted buffer. (The inserted buffers increase the static, adiabatic, and non-adiabatic energy dissipations.)

Ultimately, we examine our method on the eight bit Brent-Kung adder (which is shown in Fig.3(d)). This circuit contains 58 two-input gates as well as roughly 100 inserted buffer/inverter gates. The simulation results reveal that our technique decreases the total energy consumption of the 8-bit Brent-Kung adder by about 60% in average. It should be noted that since buffers have lower capacitance as compared to other gates, considering Eq.(3),(4) the body biasing make less energy improvement in buffers as compared to other gates. Therefore because the ratio of the inserted buffers in the Brent-Kung adder is lower than the CLA adder, it has higher energy saving (about 60%) than the 8-bit CLA adder (about 50%). Moreover, the charge recovery implementations of 8-bit Brent-Kung adder has less latency (11 phases) than the 8-bit CLA adder(18 phases which can be calculated by number of circuit's stage plus one). Nevertheless, since the major concern in this paper is the investigation of body biasing on charge recovery circuits, we are not dealing with customizing and optimizing the CLA architecture for charge recovery purpose.

V. CONCLUSIONS

In charge recovery logic styles, the use of oscillating power supplies makes the threshold voltage of pull-up transistors variable. This phenomenon degrades the charge recovery (and consequently the energy efficiency) and makes them more vulnerable to power analysis attacks. To curb this degradation, this paper proposes a body biasing scheme for PMOS transistors of charge recovery gates. In this method, by decreasing the threshold voltage of PMOS transistors dynamic power consumption (which is data dependent) decreases significantly. Therefore the energy gain and DPA-resistance of the charge recovery logic style increase considerably. We have evaluated our method on 4-bit, 8-bit, 16-bit CLA adder as well as 8-bit Brent-Kung adder by detailed SPICE simulations. Our presented simulation results show that our approach reduces the energy consumption by about 50% in average. Moreover, this technique reduces the peak power consumption in each transition and accordingly reduces data-dependent energy consumption significantly.

REFERENCES

[1] Y. He, J. Tian, X. Tan, and H. Min, "Quasi-Static Adiabatic Logic 2N-2N2P2D Family," *Electronic Letters*, vol. 42, pp. 905- 906, 2006.

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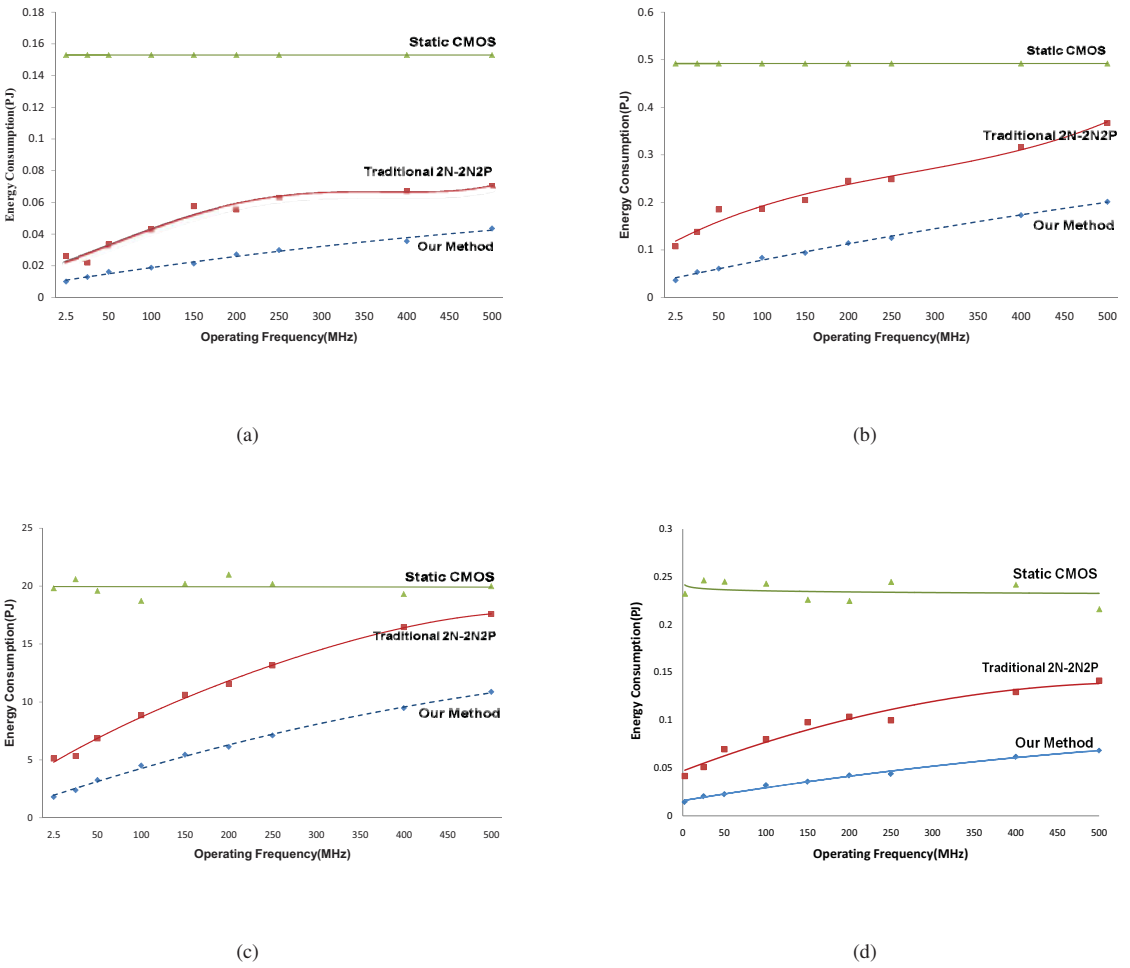


Fig. 3. Energy consumption of CLA-adder circuits, (a) 4bit CLA-adder, (a) 8bit CLA-adder, (a) 16bit CLA-adder, (a) 8bit Brent-Kung adder.

[2] S. Kim and M.C. Papaefthymiou, "True Single-Phase Adiabatic Circuitry," In *IEEE Trans. on VLSI Systems*, vol. 9, no. 1, pp. 52-63, 2001.

[3] Y. Moon and D.-K. Jeong, "An Efficient Charge Recovery Logic Circuit," In *IEEE Journal of Solid State Circuits*, vol. 31, pp. 514-522, 1996.

[4] A. Kramer, J.S. Denker, B. Flower, and J. Moroney, "2nd Order Adiabatic Computation with 2N2P and 2N-2N2P Logic Circuits", In *Proc. International Symposium on Low Power Design*, pp. 191-196, 1995.

[5] P.C. Kocher, Timing Attacks on Implementations of Diffie-Hellman, RSA, DSS, and Other Systems, In CRYPTO 96, vol. 1109 of LNCS, Springer, pp. 104-113, 1996.

[6] P.C. Kocher, J. Jaffe, and B. Jun, Differential Power Analysis, In CRYPTO 99, vol. 1666 of LNCS, Springer, pp. 388-397, 1999.

[7] M. Khatir, A. Moradi, A. Ejlali, M. T. Manzuri Shalmani, and M. Salmasizadeh, "A Secure and Low-Energy Logic Style Using Charge Recovery Approach", In *Proc. of ISLPED'08*, pp.259-264, 2008.

[8] A. Moradi, M. Khatir, M. T. Manzuri Shalmani, and M. Salmasizadeh, "Charge Recovery Logic as a Side Channel Attack Countermeasure", In *Proc. of ISQED'09*, pp.686-691, 2009.

[9] S. Mangard, E. Oswald, and T. Popp, "Power Analysis Attacks, Revealing the Secrets of Smart Cards," *Springer*, 2007, ISBN 0-387-30857-1.

[10] B. Wang and P. Mazumder, "On Optimality of Adiabatic Switching in MOS Energy-Recovery Circuit," In *Proc. of ISLPED'04*, pp. 236-239, 2004.

[11] V. Sathe, J.-Y. Chueh, and M. C. Papaefthymiou, "Energy-efficient GHz-class Charge Recovery Logic", In *IEEE J. Solid-State Circuits*, Vol. 42, No. 1, January 2007.

[12] A. Vetuli, S.D. Pascoli, and L.M. Reyneri, "Positive feedback in adiabatic logic," *Electronics Letters,* vol. 32, pp. 1867-1869, 1996.

[13] S. Narendra, D. Antoniadis, V. De, "Impact of using adaptive body bias to compensate die-to-die Vt variation on within-die Vt variation", *In Proc. of ISLPED'99*, pp. 229- 232, 1999.

[14] M.J. Deen, O. Marinov, "Effect of forward and reverse substrate biasing on low-frequencynoise in silicon PMOSFETs", In *IEEE Trans. On Electron Devices*, Vol. 49, Issue 3, pp.409 - 413, Mar 2002.

[15] F. Mac, F.-X. Standaert, J.-J. Quisquater, "Information Theoretic Evaluation of Side-Channel Resistant Logic Styles", In *CHES 2007*, vol. 4727 of LNCS, Springer, pp 427-442, September 2007.

[16] B. Gierlichs, L. Batina, P. Tuyls, and B. Preneel, "Mutual Information Analysis - A Generic Side-Channel Distinguisher", In *CHES 2008*, vol. 5154 of LNCS, Springer-Verlag, pp. 426-442, 2008.

[17] T.M. Cover, J.A. Thomas, "Elements of Information Theory", Wiley Interscience, pp.261-300, Jul 2006.

[18] Christian Piguet, "Low-Power CMOS Circuits Technology, Logic Design and CAD Tools"', CRC Press, 2006

[19] M. Bucci, L. Giancane, R.O Luzzi, A. Trifiletti, Three-Phase Dual-Rail Pre-charge Logic, In CHES 2006, vol. 4249 of LNCS, Springer, pp. 232-241, 2006.

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